ISAM 102 Workshop Bill of Materials

See `github.com/robzach/ISAM_102_workshop` for much more information and purchasing links

Bill Nace, `wnace@andrew.cmu.edu`
Robert Zacharias, `rzachari@andrew.cmu.edu`

item	part number	quantity needed per kit	unit	price per unit	consumables cost/kit	note
	total comsumables for one kit:				**$36.00**	
Electronics						
continuous rotation servo, SpringRC SM-S4303R	E1	1	each	$9.98	$9.98	
medium breadboard	E2	1	each	$6.65	$6.65	
solid-core jumper wires	E3	30	inch	$0.01	$0.18	breadboard connections; see "jumper wires" tab for details
silicone stranded wires	E4	38	inch	$0.01	$0.22	LED wires (need to flex easily)
IR proximity sensor, LTH 1550-01	E5	1	each	$0.50	$0.50	
5.6kΩ resistor	E6	1	each	$0.03	$0.03	voltage divider for IR proximity sensor's phototransistor
220Ω resistor	E7	1	each	$0.03	$0.03	current limiter for IR illuminator
3 x AA battery holder	E8	1	each	$1.51	$1.51	
AA battery	E9	3	each	$0.50	$1.50	
5mm red/green LED	E10	1	each	$0.35	$0.35	
5mm LED fresnel lens	E11	1	each	$0.51	$0.51	makes LED panel-mountable
10k potentiometer, solder lugs, w/nut and washer	E12	1	each	$1.30	$1.30	
male–male long header	E13	1	3-pin piece	$0.05	$0.05	for breadboard–> servo motor connection
rocker switch	E14	1	each	$0.40	$0.40	main power switch
Wago connector	E15	1	each	$0.53	$0.53	
male header pin	E16	1	each	$0.01	$0.01	battery pack ground
Molex crimp pin	E17	2	each	$0.23	$0.46	LED leads terminals
Raspberry Pi Pico H	E18	1	each	$5.00	$5.00	microcontroller
heat shrink tubing	E19	2	inch	$0.03	$0.06	electrical insulation for eye LED legs
spade terminal, female, 0.187" (4.75mm)	E20	1	each	$0.33	$0.33	for mating with power switch
10Ω resistor	E21	1	each	$0.03	$0.03	red/green LED current limiter
2-pin Molex housing	E22	1	each	$0.24	$0.24	LED leads terminals

Hardware/mechanical parts						
binding barrel, 1/2" length, 1/4" shaft diameter	H1	1	each	$0.35	$0.35	secures head to body
#10-24 x 3/8" screw for binding barrels	H2	1	each	$0.04	$0.04	
brass fasteners, 1" length	H3	1	each	$0.01	$0.01	
1/4" steel rod, 4-1/8" length	H4	1	each	$0.50	$0.50	cut down from 2' rods
1/4" ID x 5/8" OD flanged ball bearing	H5	1	each	$1.27	$1.27	
1/4" bushing with shoulder	H6	2	each	$0.55	$1.10	
1/4" washer	H7	1	each	$0.07	$0.07	
M3 x 15mm bolt	H9	11	each	$0.07	$0.72	
M3 x 20mm bolt	H10	6	each	$0.12	$0.70	
M3 nut	H11	12	each	$0.02	$0.25	
ziptie, 8"	H14	1	each	$0.01	$0.01	
Phillips head #2 wood screw	H16	2	each	$0.08	$0.17	
ziptie, 4"	H20	5	each	$0.01	$0.03	
1/4" wood dowel, 3-1/2" length	H21	1	each	$0.05	$0.05	cut down from 3' dowels
M3 heat-set brass insert	H22	1	each	$0.20	$0.20	
sticky mount point for ziptie	H23	1	each	$0.06	$0.06	
M3 lock nut	H24	4	each	$0.05	$0.20	
M3 washer	H25	8	each	$0.02	$0.18	
M3 x 6mm bolt	H26	4	each	$0.06	$0.23	
servo horn screw (from servo kit)	H27	1	each			
cardboard	H28	1	sq. in.			
Makerspace-manufactured parts						
3D printed parts						
1/4" shaft hub, clamping	D1	2	each			
snap-fit tail loop, small	D2	1	each			
tail rotor	D3	1	each			
gear, 36 tooth	D6	1	each			
1/4" shaft hub, no clamp	D7	1	each			
potentiometer knob	D8	4	each			
snap-fit tail loop, medium	D9	1	each			
snap-fit tail loop, large	D10	1	each			
Lasercut Parts						**cut from**
acrylic wall	L1	1	each			6mm clear acrylic
support arch	L2	1	each			6mm plywood
base	L3	1	each			6mm plywood

gear, 12 tooth	L4	1	each			1/8" two-tone acrylic
gear, 44 tooth	L6	1	each			6mm plywood
dog body	L7	1	each			6mm plywood
dog head	L8	1	each			6mm plywood
pentacam	L9	1	each			6mm plywood
shoulder wall	L13	1	each			3mm plywood
spiralcam	L14	1	each			6mm plywood
cardboard building kit	L15	1	set of pieces			single-ply cardboard
Workshop materials and tools						
total tools/materials cost for workshop of 15 teams:					**$868.34**	
Allen wrench for M3 button-head hardware	T1	60	each	$0.37	$22.20	
Phillips screwdriver, #1	T2	15	each	$0.70	$10.49	
Phillips screwdriver, #2	T3	15	each	$1.87	$28.09	
needlenose pliers	T4	15	each	$1.50	$22.50	
heat gun	T5	1	each	$253.75	$253.75	for heat shrink
Molex crimping tool	T6	1	each	$89.98	$89.98	for Molex header pins
hot glue gun w/glue	T7	3	each	$2.60	$7.80	
mallet (high-precision momentum transfer device)	T8	8	each			made from 2x4
plastic bags for parts organization	T9	90	each	$0.07	$6.30	
cardboard knife, Canary	T12	2	each	$7.79	$15.58	
document camera, VZ-R	T13	1	each	$280.10	$280.10	
generic crimping tool	T14	1	each	$6.98	$6.98	
printed name tags	T15	1	set			generated from CSV by script
wire stripper, Ideal Stripmaster	T16	1	each	$75.19	$75.19	
generic wire stripper	T17	3	each	$3.38	$10.14	for making breadboard wires
diagonal flush cutters	T18	15	each	$1.89	$28.35	
markers	T19	15	each	$0.56	$8.40	
3d printed wire-bending jigs	T20	3	each			for making breadboard wires
USB-C to USB-B Micro	T22	1	each	$2.50	$2.50	for programming Raspberry Pi Picos, and connecting computer to doc cam

Bolt measurement reference

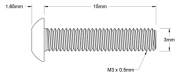
Socket-head bolts: measure length from underside of bolt head.

Images stolen shamelessly from McMaster-Carr.

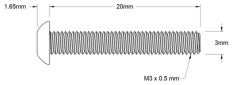
M3 × 6mm bolt



M3 × 15mm bolt



M3 × 20mm bolt



System schematic

Drawn using our customized ioRef flavor of `draw.io` diagramming tool; access the customized version via `draw.ioref.org`

